Exact and Efficient Unlearning for Language Models

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Abstract

Language Models (LMs) can memorize information from their training datasets, which might include private or copyrighted material. This raises legal and ethical concerns, as LMs can unintentionally reproduce this sensitive information during inference. In this respect, legal frameworks require that model owners forget specific data from LMs upon request from the referred individuals or data owners.

Machine unlearning aims to achieve data forgetting with lower computational cost than retraining LMs from scratch without the undesired data. However, most unlearning methods are *approximate* and lack the formal guarantees of forgetting required for legal compliance. Meanwhile, the few existing frameworks that provide *exact* unlearning guarantees are impractical for LMs due to their high computational cost.

In this paper, we introduce a new framework for exact unlearning that optimizes data distribution during training to enable more efficient enforcement of unlearning requests. Our approach assigns a forgetting priority score to each training sample based on its recency and/or sensitivity, ensuring that samples more likely to be forgotten are placed in more easily and quickly removable data partitions.

We evaluate our framework on BERT-based models for two critical use cases: copyright and privacy protection. Our experiments demonstrate that our framework significantly reduces unlearning times compared to existing exact unlearning methods while preserving model utility. Further, compared to four approximate unlearning methods, our approach achieves a superior balance of efficiency, forgetting quality, and utility.

Keywords: Machine Unlearning, Privacy Protection, Copyright Content Protection, Language Models

1 Introduction

Large Models (LMs) have demonstrated remarkable capabilities across a wide range of Natural Language Processing (NLP) tasks, from text generation and translation to question answering and code completion [4, 5]. These models have achieved near-human or superhuman performance on various benchmarks, including reasoning tasks and professional examinations. This unprecedented success can be attributed to the availability of vast amounts of training data, primarily sourced through web scraping, which often includes content created by individuals, organizations, and businesses.

Despite their impressive performance across various NLP tasks, LMs have also raised serious concerns about data privacy and ownership. Recent studies have shown that LMs can memorize and reproduce sensitive training data, including personal information and copyrighted content [10, 11]. For example, researchers have shown that models can leak personal email addresses and phone numbers [12]. Furthermore, studies have highlighted that LMs can unintentionally generate text that closely resembles copyrighted works, including books, articles, and software code, which may result in potential copyright violations [7].

To mitigate these risks, regulatory frameworks have evolved to enforce data protection and intellectual property rights. The European Union's General Data Protection Regulation (GDPR) and AI Act impose strict requirements for data privacy and AI governance [16]. Similarly, the California Consumer Privacy Act (CCPA) in the United States provides protections for personal information. Copyright laws such as the Digital Millennium Copyright Act (DMCA) and the EU Copyright Directive further safeguard intellectual property in AI-generated content.

These legal frameworks establish the right to digital forgetting, requiring mechanisms to remove specific data from machine learning (ML) models upon request. A straightforward but computationally pro-

hibitive approach involves retraining models from scratch after removing the designated data [4]. This method demands extensive resources and is impractical for large-scale models.

To address these challenges, researchers have explored machine unlearning techniques that selectively remove specific data without requiring full retraining. Exact unlearning methods provide strong guarantees of data removal but often incur high computational costs [17]. In contrast, approximate unlearning techniques offer more practical solutions but may lack formal assurances of complete data removal [19]. Developing efficient and scalable unlearning techniques remains essential for ensuring compliance with privacy regulations and copyright laws while maintaining model performance.

1.1 Contributions and plan

In this paper, we address the critical challenges of unlearning in large models (LM) by exploring how the SISA framework can be adapted for textual data. Unlike existing unlearning techniques, which either require expensive full retraining or rely on suboptimal data partitioning strategies, our approach investigates data distribution strategies across shards and slices to facilitate faster and more effective unlearning. Our focus is on optimizing unlearning efficiency in two key areas, which are the primary concerns of most legal frameworks: copyright protection and private information safeguarding.

There are currently no exact unlearning methods specifically designed for large models; existing approaches primarily target standard machine learning models with classic training data. While previous SISA implementations have focused on image data [17] or structured datasets [18], we study how similar principles can be applied to textual data and large models. By refining data partitioning strategies, we aim to improve unlearning efficiency while maintaining model utility.

To evaluate our approach, we benchmark it against four alternative approximate unlearning methods and conduct extensive experiments under various unlearning scenarios. Our results provide insights into the trade-offs between forgetting guarantees, computational cost, and model performance.

The rest of this paper is structured as follows: Section 2 reviews related work on machine unlearning and existing SISA-based approaches. Section 3 details our methodology and proposed framework. Section 4 describes the experimental setup, including datasets, models, and evaluation metrics. Section 5 presents our results and analysis. Finally, Section 6 concludes the paper with a discussion of limitations and future research directions.

2 Background and related work

Machine unlearning (MU) is an emerging area of research that focuses on efficiently removing specific data points or features from trained machine learning (ML) models without compromising their overall performance. The core aim of MU is to ensure that models remain adaptable and can exclude sensitive or irrelevant information, thus preserving privacy, reducing bias, and improving fairness. For example, in scenarios where a model has been trained on a large dataset containing copyrighted content, MU can be used to remove the influence of such content without retraining the model from scratch. This is crucial in compliance with privacy regulations and for maintaining the ethical integrity of models deployed in real-world applications [43].

The concept of machine unlearning was first introduced by Cao et al. [43], who proposed a "forgetting system" that helped address the challenge of removing specific data points from a trained model. Their approach reformulated traditional learning algorithms into a summation form, which allowed for the exclusion of harmful data without degrading model performance. One of the key motivations for the development of machine unlearning techniques is the growing demand for privacy compliance, as driven by regulatory frameworks such as the General Data Protection Regulation (GDPR) [45], the California Consumer Privacy Act (CCPA) [46], and the California Privacy Rights Act (CPRA) [47]. These regulations mandate not only the deletion of user data from storage but also its removal from any models that have been trained using that data.

The machine unlearning pipeline consists of three main stages: training, inference, and unlearning. First, a model $\mathbf{w}^*$ is trained on the initial dataset $\mathcal{D}_{init}$ containing n_{init} data points. The dataset is represented as $\mathcal{D} \subset \mathcal{D}_{init}$, where $\mathcal{D}$ may be a subset after deletions.

The fitness of the model parameters $\mathbf{w}$ is measured by an objective function L, which is given by:

$$L(\mathbf{w}, \mathcal{D}) = \frac{1}{|\mathcal{D}|} \sum_{i \in \mathcal{D}} \ell(\mathbf{w}^T \mathbf{x}_i, y_i) + \frac{\lambda}{2} \|\mathbf{w}\|_2^2$$

where ℓ is the binary cross-entropy loss and the second term represents ridge regularization. The model is trained using stochastic gradient descent (SGD), updating the model as follows:

$$\mathbf{w}_{t+1} = \mathbf{w}_t - \eta_t \nabla L(\mathbf{w}_t, \mathcal{D})$$

where η_t is the learning rate.

During the inference stage, the trained model $\mathbf{w}^*$ is used to predict labels for new data points:

$$\hat{y}_i = \text{Infer}(\mathbf{w}^*, \mathbf{x}_i)$$

The unlearning stage begins when part of the data $\mathcal{D}_m$ is deleted. The updated model $\mathbf{w}_u$ is generated by adjusting the parameters to "forget" the influence of the deleted data. This is done by applying an unlearning algorithm that updates the model using the remaining data $\mathcal{D} \setminus \mathcal{D}_m$.

The updated model is then evaluated to ensure it meets the desired performance. If the performance is satisfactory, the updated model is used for further inference. If not, the process may return to the training stage for a full retraining on the remaining data.

This pipeline in Figure 1 provides a computationally efficient way to remove specific data points without the need for complete retraining, which is crucial for privacy compliance and fairness in machine learning models.

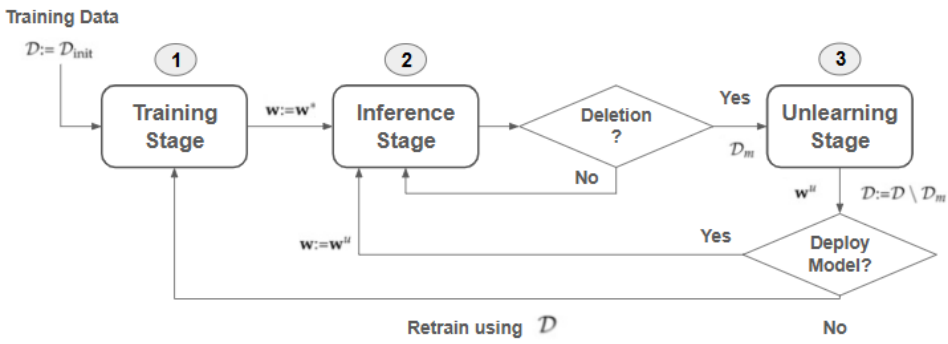

Figure 1: Machine Unlearning Pipeline

2.1 Exact vs. Approximate Unlearning

As machine learning models continue to grow in complexity, the ability to unlearn specific data has become an urgent challenge. With models learning from vast amounts of data, reducing data consumption while maintaining performance [54] has become increasingly difficult. Unlearning began to gain serious attention, especially as models started using sensitive, personal information for unintended purposes, raising concerns about privacy and security. This is why it became imperative to develop mechanisms for **exact unlearning**, ensuring that specific data can be completely forgotten.

Exact Unlearning Achieving 100% exact unlearning may seem straightforward in theory, but it is extremely difficult to implement in practice. The most straightforward method for unlearning is retraining from scratch, where the model is retrained entirely from the beginning without including the data that needs to be forgotten. This method guarantees that no unwanted and sensitive data remains, but it is highly resource-intensive, especially with large-scale models like LLMs (Large Language Models).

Another approach considered when it comes to exact unlearning is SISA (Sharded, Isolated, Sliced, and Aggregated), which, while not as exhaustive as retraining from scratch, provides a more efficient method for unlearning. SISA divides data into shards and slices, retraining only the parts that must be forgotten upon forget requests. However, even this method tends to be limited when dealing with large datasets and complex models, as it does not fully address issues with overlapping data dependencies.

Approximate Unlearning Approximate Unlearning More recently, there has been a strong focus on approximate unlearning, which seeks to remove specific data more quickly and with fewer computational resources compared to exact unlearning methods. These approximate methods aim to reduce the influence of sensitive data in the model without requiring a full retraining process.

The TOFU paper [20] uses four distinct approximate unlearning methods, which can be briefly summarized as follows:

- **Gradient Ascent:** Optimizes the model by iteratively updating parameters in the direction that maximally reduces the impact of unlearned data. The update rule is given by:

$$\theta_{t+1} = \theta_t + \eta \nabla_\theta L(\theta_t)$$

  where θ_t are the model parameters at time t, η is the learning rate, and $L(\theta_t)$ is the loss function with respect to the model parameters.

- **Gradient Difference:** Calculates and applies the difference in gradients before and after unlearning to adjust the model efficiently. The gradient update can be written as:

$$\Delta\theta = \nabla_\theta L(\theta_{\text{after}}) - \nabla_\theta L(\theta_{\text{before}})$$

  where $\nabla_\theta L(\theta_{\text{after}})$ and $\nabla_\theta L(\theta_{\text{before}})$ are the gradients computed before and after unlearning, respectively.

- **KL Minimization:** Minimizes the Kullback-Leibler[1] divergence to ensure the model distribution closely aligns with the desired state after unlearning. The KL divergence between two probability distributions $P(\theta)$ and $Q(\theta)$ is given by:

$$D_{\text{KL}}(P \parallel Q) = \int P(\theta) \log \left(\frac{P(\theta)}{Q(\theta)} \right) d\theta$$

  where $P(\theta)$ is the original model distribution and $Q(\theta)$ is the distribution after unlearning.

- **Preference Optimization:** Adjusts the model parameters to prioritize outcomes that are less influenced by the unlearned data. The optimization objective can be written as:

$$\theta^* = \arg\min_\theta \left(\mathbb{E}[\mathcal{L}(\theta)] + \lambda \|\theta - \theta_{\text{prior}}\|^2 \right)$$

  where $\mathcal{L}(\theta)$ is the loss function, θ_{prior} is the previous model state, and λ is a regularization parameter that balances the influence of the prior model state.

Each of these methods has its strengths and weaknesses in terms of scalability and resource efficiency. While approximate unlearning techniques are less resource-intensive and more practical for large models like LLMs, they do not guarantee complete removal of the sensitive information. Instead, they offer a partial or temporary mitigation of the impact of the forgotten data [56].

Therefore, while exact unlearning methods guarantee full compliance with privacy laws (such as GDPR and CCPA), they are generally not suitable for large models like LLMs, which are frequently subject to unlearning requests due to their extensive use of personal data. On the other hand, approximate unlearning methods are more commonly used in practice for operational efficiency but do not address the underlying legal concerns surrounding complete data removal.

Table 1: Comparison of Exact and Approximate Unlearning Methods

Aspect	**Exact Unlearning (e.g., SISA)**	**Approximate Unlearning (e.g., Gradient Ascent)**
Data Removal Guarantee	100% (complete erasure of data)	Low (no guarantee, data influence mitigation)
Legal Standards	High (full compliance with privacy laws)	Low (legal compliance not guaranteed)
Computational Cost	High (requires retraining or sharding for LLMs)	Low (more efficient and faster)

Table 1: Comparison of Exact and Approximate Unlearning Methods

Aspect	Exact Unlearning (e.g., SISA)	Approximate Unlearning (e.g., Gradient Ascent)
Scalability	Moderate (challenging to scale it up)	High (reduced computational cost)
Privacy Preservation	High (full data erasure and no data leakage)	Low (privacy leakage and incomplete forgetting)
Verification	Guaranteed and verifiable	Difficulty to verify

2.2 Legal Frameworks for Unlearning

Unlearning methods are primarily motivated by explicit legal requirements related to privacy and copyright. Legal frameworks such as the General Data Protection Regulation (GDPR) and the California Consumer Privacy Act (CCPA) establish clear obligations for data removal upon request. For instance, GDPR's "Right to be Forgotten" (Article 17) grants individuals the right to request the deletion of their personal data, while the CCPA provides similar protections, requiring organizations to erase consumer data upon request. These regulations emphasize the need for legally compliant unlearning mechanisms that can verifiably remove specific data points from machine learning models.

In this work, we focus on the two primary types of legally recognized unlearning requests: **privacy-related** and **copyright-related**. Privacy-related requests involve the removal of personally identifiable information (PII), such as user biographies or medical records, while copyright-related requests require the deletion of proprietary content, such as book excerpts or media assets [54, 55]. These requests necessitate exact unlearning to ensure full compliance, as approximate methods do not provide the necessary guarantees for verifiable data removal [54].

Given these constraints, existing unlearning approaches must be assessed in terms of their ability to meet legal requirements while remaining practical for large-scale models. Methods like SISA (Sharded, Isolated, Sliced, and Aggregated) offer strong guarantees for exact unlearning but face scalability challenges, particularly when applied to large language models (LLMs). This trade-off has motivated ongoing research into hybrid approaches that seek to balance computational feasibility with strict compliance.

2.3 Motivation for Digital Forgetting

The need for digital forgetting in machine learning, particularly in Large Language Models (LLMs), is driven by several key factors:

1. **Privacy:** Protecting individuals' privacy is paramount, especially in the context of machine learning models that are trained on vast datasets. These models may unintentionally include private or Personally Identifiable Information (PII), which could be exposed if not properly managed. Misconfigured services can result in such sensitive data being indexed and accessible, highlighting the necessity for mechanisms to remove this information from models.
2. **Copyright Protection:** To prevent the leakage of verbatim text from copyrighted works, digital forgetting mechanisms are essential. While privacy laws protect personal information, copyright laws safeguard the specific form of expression, making it crucial to prevent the unauthorized reproduction of protected content.
3. **Model Robustness and Ethical Considerations:** Machine learning models must be designed to avoid perpetuating harmful biases or generating offensive content. Models trained on biased datasets risk amplifying stereotypes and producing outputs that are misaligned with ethical standards. Ensuring models are robust and ethical is an important motivation for digital forgetting, as it helps mitigate harmful impacts on society.
4. **Legal and Ethical Compliance:** Digital forgetting is also driven by legal requirements, such as the General Data Protection Regulation (GDPR), which mandates the ability to erase personal data upon request. These regulations ensure that digital forgetting measures are implemented to comply with privacy laws and protect individuals' rights.
5. **Alignment with Human Values:** Machine learning models should align with broader ethical val-

[1] *The Kullback-Leibler divergence is a measure of how one probability distribution diverges from a second, expected probability distribution. It is often used in machine learning to quantify the difference between two probability distributions.*

ues, including fairness and non-discrimination. Adherence to guidelines such as the EU Guidelines for Trustworthy AI necessitates identifying and removing sources of harmful behavior in models. Digital forgetting plays a critical role in ensuring that models reflect and respect human values.

Together, these motivations underscore the importance of digital forgetting as a means to ensure privacy, compliance, and ethical standards in the development and deployment of machine learning models.

2.4 Related Work

2.4.1 Overview of Existing Methods: SISA and its variants

SISA, which stands for Sharded, Isolated, Sliced, and Aggregated training, is an unlearning framework introduced by Bourtoule et al. (2021) that guarantees exact unlearning as mentioned in the previous section. The framework operates by partitioning the training dataset into multiple disjoint shards, with each shard further divided into several slices. A separate model is trained for each shard, processing the data slice by slice and saving checkpoints at the end of each slice.

The training methodology involves creating multiple instances of the base model, each trained on an exclusive subset of the entire dataset, referred to as a slice. These individually trained models are called constituent models. Within each slice, the data is further subdivided into smaller units to enable progressive learning, with the model state saved periodically at checkpoints. Each model learns only from its assigned slice, ensuring that training is isolated and systematic. Once all constituent models are trained, their outputs are combined to create a comprehensive model. During inference, predictions from all the models are aggregated to produce the final result. Figure 2 illustrates this methodology visually.

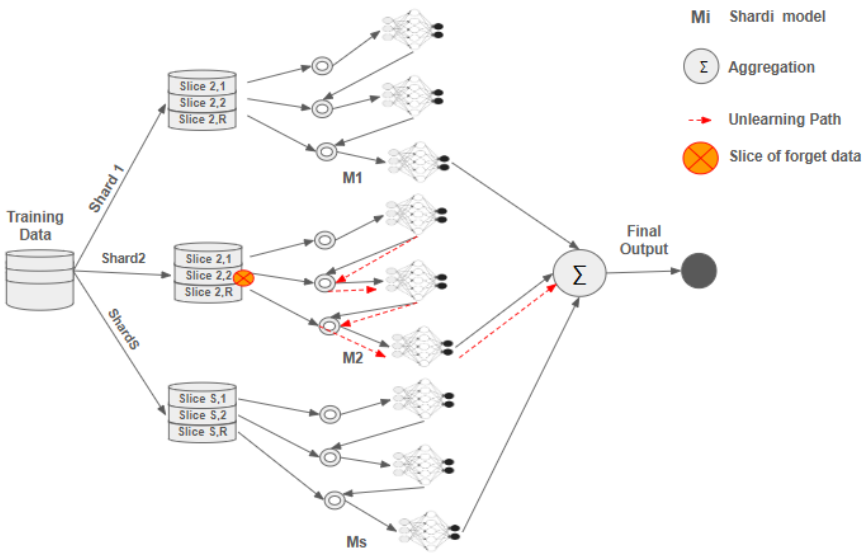

Figure 2: SISA Framework

When an unlearning request is submitted, the framework identifies the specific shard and slice containing the data point to be removed. The data point is deleted from the relevant slice, and the model is retrained starting from the last unaffected checkpoint [66]. This approach ensures the model can effectively "forget" the specified data without compromising its overall functionality.

Building on the foundational principles of SISA [61], researchers have proposed various variants tailored to improve efficiency in handling unlearning tasks for large language models (LLMs), particularly addressing the computational challenges inherent in the original framework. We will be explaining two variants that are the most closely aligned with the original SISA:

1. **SISA-FC**: This variant starts from a base model that is pre-trained on a generic text corpus and then adds fully connected (FC) layers on top. Only the parameters from these added layers are updated during retraining, which minimizes the overall retraining time as backpropagation of gradients occurs only in the final layers [62]. However, this approach may significantly affect the utility of the model compared to fine-tuning the entire model.
2. **SISA-A**: This variant employs the parameter-efficient adapter method, which injects learnable

modules into the encoder blocks of the transformer [67]. SISA-A aims to preserve the classification utility of the resulting model better than SISA-FC, but it requires more computational and storage costs. Like SISA, these proposed methods do not provide exact unlearning guarantees, and their retain accuracy is lower than that of the original SISA [61].

2.4.2 Limitations of SISA for Large Language Models

The variations of SISA (Sharded, Isolated, Sliced, and Aggregated training) were introduced to address some of the inefficiencies in the original framework [61]. However, these adaptations fail to fully resolve the core limitations of SISA, particularly when applied to large-scale language models (LLMs). The key issues can be summarized as follows:

1. **High Computational and Memory Cost**: SISA incurs significant computational and memory overhead, making it impractical for many LLM applications. These resource demands arise from processes such as model training, checkpoint saving, retraining, and inference [62].
2. **Fairness Concerns**: Research indicates that SISA can compromise fairness in LLMs. While it ensures exact unlearning guarantees, it may unintentionally introduce or amplify biases within the model [64].
3. **Complexity of Model Management**: Implementing SISA necessitates managing multiple models and checkpoints for different data shards, which increases the system's complexity and diminishes its practicality for large-scale deployments [61].

These limitations underscore that, although SISA provides a robust theoretical foundation for unlearning, its practical adoption in large-scale language models remains challenging due to resource-intensive operations and fairness-related concerns.

Table 2: Comparison of SISA Variants and Their Limitations

Method	Forgetting	Retaining	Runtime	Limitations
SISA	Exact	Medium-High	Significant	High computational costs; unsuitable for LLMs; requires large storage for checkpoints.
SISA-FC	Weak/None	Low	Low	Impacts model utility; performance inferior to full fine-tuning.
SISA-A	Weak/None	Medium	Medium	Higher costs than SISA-FC; no exact unlearning; accuracy lower than SISA.

3 Methodology

3.1 Motivation and Objectives

The primary challenge in machine unlearning for large language models (LLMs) lies in achieving **exact unlearning** while minimizing computational costs. This is especially critical in contexts such as **privacy** and **copyright-related forgetting requests**, where legal compliance requires formal guarantees of data removal. In these scenarios, failure to ensure exact unlearning could lead to non-compliance with privacy regulations (e.g., GDPR) and intellectual property rights, resulting in significant legal repercussions.

When it comes to exact unlearning, we can detail these two general approaches:

- **Retraining from scratch**: This approach ensures exact data removal by fully retraining the model f_θ after deleting the requested data $\mathcal{D}_m$ from the initial dataset $\mathcal{D}_{\text{init}}$. While it guarantees compliance with privacy and copyright regulations, it is computationally prohibitive for large-scale LLMs due to their billions of parameters, high energy consumption, and significant carbon footprint [68, 69].
- **SISA training** [61]: SISA reduces retraining costs by partitioning $\mathcal{D}_{\text{init}}$ into *shards* and *slices*, retraining only the affected partitions upon a forgetting request. However, due to the random data distribution, multiple shards may still require retraining, limiting efficiency gains. This randomness can lead to unnecessary computations, making SISA less optimal for large-scale deployments [61].

Our proposed methodology aims to further optimize the SISA framework by **better organizing the data partitions** and **strategically distributing the data points** across shards and slices. This

improvement seeks to reduce computational costs while maintaining **exact unlearning** guarantees.

Problem Formulation In the SISA framework, the data is partitioned into **shards** S and further subdivided into **slices** R. However, due to the aggregation of checkpoints across these slices, a deletion request for a data point in one slice may force the retraining of multiple slices. For example, we will consider a case where we have 3 slices and 5 shards. If the data point to be unlearned, $\mathcal{D}_m$, falls in slice R_1, then all slices that share its checkpoint (i.e., R_1, R_2, and R_3) must be retrained. Similarly, if $\mathcal{D}_m$ is in R_2, then both R_2 and R_3 need retraining; and if it is in R_3, only R_3 requires retraining.

This dependency can be formalized as follows:

$$\text{Unlearn}(\mathcal{D}_m) \Rightarrow \begin{cases} \text{Retrain}(R_1 \cup R_2 \cup R_3) & \text{if } \mathcal{D}_m \in R_1, \\ \text{Retrain}(R_2 \cup R_3) & \text{if } \mathcal{D}_m \in R_2, \\ \text{Retrain}(R_3) & \text{if } \mathcal{D}_m \in R_3. \end{cases}$$

Figure 3 illustrates this dependency. In the diagram, the slices R_1, R_2, and R_3 are arranged vertically with their respective checkpoints C_1, C_2, and C_3.

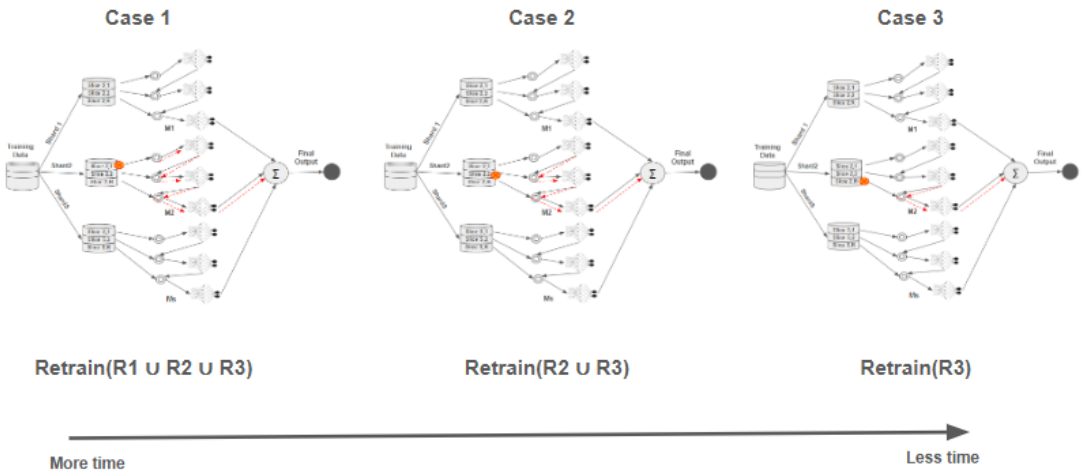

Figure 3: SISA Use Cases - If the data to forget falls in the slice 1: R1, R2 and R3 slices needs to be retrained

This example highlights that the random distribution of data among slices and shards in SISA can lead to redundant retraining. For instance, a deletion in an earlier slice (e.g., R_1) forces retraining of all subsequent slices, thereby increasing computational overhead and negating the efficiency gains intended by the SISA approach.

In addition to the dependency induced by the slicing mechanism, it is important to note that the ordering of shards also significantly impacts retraining efficiency. In SISA, shards are typically arranged chronologically, meaning that later shards (i.e., those created more recently) tend to require considerably more retraining time than earlier ones. Thus, if a deletion request targets data within a later shard, the retraining process becomes even more costly—not only due to the need to retrain multiple slices (as described previously) but also because the affected shard's position implies a higher computational overhead.

This additional factor further illustrates the drawbacks of the random data distribution strategy employed by SISA. Not only can a deletion in an earlier slice (e.g., R_1) force the retraining of several subsequent slices due to shared checkpoints, but if the deleted data resides in one of the later shards, the retraining process is compounded by the inherently longer retraining times associated with those shards. This dual dependency on both slice position and shard ordering underscores the complexity and potential inefficiencies in the SISA framework when handling deletion requests.

3.2 Proposed Framework

Our proposed framework, presented in Figure 4, adds a data pre-processing stage before the SISA-based sharding and slicing stages. Next, we train on these shards and slices while saving checkpoints, and

finally, we execute the unlearning process. By prioritizing structured and balanced partitioning, we enhance the precision and efficiency of unlearning. This approach not only overcomes the limitations of random distribution in the original methodology but also ensures that models remain both legally compliant and computationally efficient.

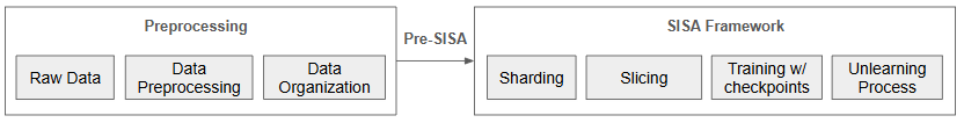

Figure 4: Proposed Framework Pipeline

Data Preprocessing

Forgetting Priority Score

In our framework, we aim to compute a *forget priority score* $F(i)$ for each data sample i that captures two dimensions: *recency* and *sensitivity*. Recency indicates how recent the content is, while sensitivity quantifies the potential copyright or privacy risk. By incorporating recency, we prioritize more recent content, which is more likely to be subject to forgetting requests. Sensitivity, on the other hand, allows us to measure how closely a data sample may be associated with potential risks, whether from copyright infringement or privacy violations.

Organizing data according to the likelihood of forgetting enhances the efficiency of the unlearning process while providing precise guarantees. This approach ensures that the most sensitive or time-sensitive data is removed first. Such an arrangement is particularly beneficial in situations where a precise and efficient forgetting strategy is essential, such as complying with data protection regulations or reducing the risks of legal penalties associated with outdated or sensitive content. The overall score is defined as follows:

$$F(i) = \alpha R\mathrm{r}(i) + \beta S\mathrm{s}(i), \tag{1}$$

where α and β are weights that adjust the relative importance of recency $R\mathrm{r}(i)$ and sensitivity $S\mathrm{s}(i)$, respectively. The sensitivity score $S\mathrm{s}(i)$ itself is a weighted sum of several individual sensitivity features:

$$S\mathrm{s}(i) = \sum_{k=1}^{N} \gamma_k S\mathrm{s}_k(i), \tag{2}$$

with $S\mathrm{s}_k(i)$ denoting the k sensitivity metric and γ_k its corresponding weight.

For **copyright**, recency is determined by the publication year. Given that works older than 70 years are generally in the public domain [70], we define the recency score as:

$$R\mathrm{r}(i) = \begin{cases} \dfrac{\mathrm{Year}(i)}{\max\{\mathrm{Year}\}}, & \text{if } \max\{\mathrm{Year}\} - \mathrm{Year}(i) \leq 70, \\ 0, & \text{otherwise.} \end{cases} \tag{3}$$

This definition ensures that only recent works (within the copyright period) receive a nonzero score, normalized between 0 and 1.

For **privacy**, and in the absence of an established legal time frame for personal data, we approximate recency by using the individuals' birth years. We assume that younger individuals are more likely to request data removal than older (or deceased) individuals. Thus, we define the privacy recency score as:

$$Rr(i) = \frac{\text{Birth Year}(i)}{\max\{\text{Birth Year}\}}. \tag{4}$$

This formulation assigns a higher recency score to data associated with more recent birth years.

The **sensitivity score** $Ss(i)$ reflects how prone a data sample is to copyright infringement or privacy concerns. We consider different sets of sensitivity features depending on the context.

For **copyright-related** forgetting, our experiments incorporate four key sensitivity features: **Textual**

Similarity $Ss_1(i)$: This feature quantifies the word-for-word similarity between a given data sample and other copyright-protected content. Mathematically, it is expressed as:

$$Ss_1(i) = \frac{\text{Similarity}(i)}{\max\{\text{Similarity}\}}, \tag{5}$$

where Similarity(i) measures the direct textual overlap. A higher value of $Ss_1(i)$ indicates that the data sample is more similar to other protected texts.

Snippet Length $Ss_2(i)$: Longer snippets are likely to contain more substantial content that might be copyrighted. We normalize the average length of the data sample by the maximum data sample length observed:

$$Ss_2(i) = \frac{\text{AvgLength}(i)}{\max\{\text{Length}\}}. \tag{6}$$

Theme Sensitivity $Ss_3(i)$: The theme sensitivity score $Ss_3(i)$ The theme sensitivity score measures the similarity between snippet i and the other snippets in the same temporal cluster in terms of their thematic content. It is normalized as:

$$Ss_3(i) = \frac{\text{ThemeMatch}(i)}{\max\{\text{ThemeMatch}\}}, \tag{7}$$

where ThemeMatch(i) is a metric that compares the thematic content of the data sample with that of other snippets in the same temporal cluster.

Semantic Similarity $Ss_4(i)$: This feature evaluates the similarity in meaning (as opposed to direct textual matching) between a data sample and other copyrighted content. Even if the wording differs, if the underlying message is similar, the semantic similarity score will be high. It is defined as:

$$Ss_4(i) = \frac{\text{SemanticSimilarity}(i)}{\max\{\text{SemanticSimilarity}\}}. \tag{8}$$

The overall sensitivity score for copyright is then given by:

$$Ss_{\text{copyright}}(i) = \gamma_1 Ss_1(i) + \gamma_2 Ss_2(i) + \gamma_3 S_3(i) + \gamma_4 Ss_4(i), \tag{9}$$

where γ_1, γ_2, γ_3, and γ_4 are weights that determine the relative importance of each feature.

For privacy, the sensitivity score is based on two primary features:

Private Information Length $Ss_1(i)$: As in the copyright context, longer data samples may include more personal information, so we define:

$$Ss_1(i) = \frac{\text{AvgLength}(i)}{\max\{\text{Length}\}}. \tag{10}$$

Density of Private Information $Ss_2(i)$: This feature measures the concentration of sensitive personal data (such as names, addresses, etc.) within a data sample. It is quantified by:

$$Ss_2(i) = \frac{\text{PrivateDataDensity}(i)}{\max\{\text{PrivateDataDensity}\}}, \tag{11}$$

where PrivateDataDensity(i) is computed as the ratio of recognized private information to the overall length of the snippet.

Thus, the overall sensitivity score for privacy is:

$$Ss_{\text{privacy}}(i) = \gamma_1 Ss_1(i) + \gamma_2 Ss_2(i), \tag{12}$$

with γ_1 and γ_2 as the corresponding weights.

The final score $F(i)$ is a combination of the recency score and the aggregate sensitivity score:

$$F(i) = \alpha Rr(i) + \beta Ss(i),$$

with

$$Ss(i) = \sum_{k=1}^{N} \gamma_k Ss_k(i),$$

where $N = 4$ for copyright contexts and $N = 2$ for privacy contexts. For instance, in the copyright scenario:

$$F_{\text{copyright}}(i) = \alpha Rr(i) + \beta \left(\gamma_1 Ss_1(i) + \gamma_2 Ss_2(i) + \gamma_3 Ss_3(i) + \gamma_4 Ss_4(i)\right).$$

In the privacy scenario, the final score becomes:

$$F_{\text{privacy}}(i) = \alpha Rr(i) + \beta \left(\gamma_1 Ss_1(i) + \gamma_2 Ss_2(i)\right).$$

Data Organization

By utilizing the *forget priority score* $F(i)$, we can efficiently organize the data before the sharding and slicing processes. The key idea is to structure the data in a way that follows a clear, logical order, considering both recency and sensitivity. This organization is crucial, as we anticipate receiving user forget requests that align with the features we've incorporated into the ranking. For example, we can assume that more recent content is more likely to be requested for removal than older data. Our approach strategically organizes the data based on its likelihood of being subject to forget requests in the future.

In the data organization step, we prioritize recency by arranging the data in **descending order**, with the most recent years placed at the top and progressively less recent content at the bottom. After this initial sorting by year, we use additional features to further refine the organization, ensuring that the data is organized in a way that aligns with its likelihood of being forgotten based on both recency and sensitivity.

We define each data sample as X_i, where i indexes the samples. The data is organized based on recency (year) and sensitivity. To emphasize the importance of recency over sensitivity, we consider the following:

$$\alpha > \beta \quad \text{and} \quad \text{year}_1 > \text{year}_2 > \text{year}_3 > \text{year}_4 > \cdots > \text{Oldest Year}$$

Sample	**Year ↑**	**Recency Score ↑**	**Sensitivity Score ↑**	**F(I) ↑**
X_1	$year_1$	$Rr(1)$	$Ss(1)$	$F(1)$
X_2	$year_2$	$Rr(2)$	$Ss(2)$	$F(2)$
X_3	$year_3$	$Rr(3)$	$Ss(3)$	$F(3)$
X_4	$year_4$	$Rr(4)$	$Ss(4)$	$F(4)$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
X_N	Oldest Year	$Rr(N)$	$Ss(N)$	$F(N)$

Table 3: Illustration Example of Recency Score, Sensitivity Score and their respective Forgetting Priority Score

Sharding and Slicing

In Section 3.1, we discussed how random data distribution can scatter samples arbitrarily across shards and slices, increasing computational costs—especially when data requiring deletion is stored in slices that demand extensive retraining. In contrast, our method pre-organizes data based on recency and sensitivity, enabling a personalized sharding and slicing strategy that minimizes resource consumption.

We consider data samples X_i ordered by recency and sensitivity as:

$$X_1, X_2, \ldots, X_N = S(\{X_i\}).$$

Our objective is to partition these samples into S shards, each subdivided into R slices, while satisfying the following criteria:

- **Criterion 1:** Within each shard, the most recent data is stored in the final slices.
- **Criterion 2:** Across shards, recent data is prioritized in the earliest shards.

Each shard S_j (for $j = 1, 2, \ldots, S$) is defined as:

$$S_j = \{s_1^j,\, s_2^j,\, \ldots,\, s_R^j\},$$

where s_1^j contains the oldest data and s_R^j the most recent data in that shard.

We define a mapping function D to assign each data sample X_i to a specific shard and slice:

$$D(X_i) = \big(S_{g(i)},\, s_{f(i)}^{g(i)}\big),$$

where $g(i)$ allocates recent data to the earliest shards and $f(i)$ places it in the final slices within the assigned shard.

Furthermore, to ensure a balanced distribution, each slice must contain an equal number of data samples. If the total number of samples is

$$N = \sum_{i=1}^{N} N_i$$

and there are $S \times R$ slices, then each slice holds exactly

$$n = \frac{N}{S \times R}$$

samples (with rounding adjustments applied as needed). Consequently, the distribution is controlled by the capacity of each slice rather than by manually selecting which years to place where. Nonetheless, our strategy guarantees that data from the most recent years is always positioned in the final slices of each shared.

Slicing Logic Let N_1 represent the total number of data samples from the most recent year which is $year_1$ as represented in the table X. We assume that:

$$\text{Total number of slices} = 3, \quad \text{Total number of shards} = 5.$$

Sample	**Year ↑**	**Recency Score ↑**	**Sensitivity Score ↑**	**Data Samples**
X_1	$year1$	$Rr(1)$	$Ss(1)$	N_1
X_2	$year2$	$Rr(2)$	$Ss(2)$	N_2
X_3	$year3$	$Rr(3)$	$Ss(3)$	N_3
X_4	$year4$	$Rr(4)$	$Ss(4)$	N_4
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
X_N	Oldest Year	$Rr(N)$	$Ss(N)$	N_N

Table 4: Illustration Example of Data Samples, Recency Score and Sensitivity Score

The distribution process is as follows:

$$\text{If } N_1 > n, \text{ then:}$$

$$\begin{cases} \text{Move data into slice } R_3 \text{ of shard } S_1, \text{ if } N_1 > n. \\ \text{If } N_1 > (n - N_1) \text{ in } S_1, \text{ move to slice } R_3 \text{ in shard } S_2 \text{ and repeat the process.} \\ \text{Repeat this process for each shard until } N_1 = 0, \text{ then move to } N_2. \\ \text{Continue the process for subsequent datasets } N_3, N_4, \ldots \text{ until all data is distributed.} \end{cases}$$

Thus, slice R_3 in shard S_1 may initially contain data samples from year_1, while slice R_3 in shard S_2 will contain data samples from $year_1$ and $year_2$ or only $year_1$ if $N_1 > n$, and so on.

$$\text{If at any point } N_1 < n, \text{ then:}$$

$$\begin{cases} \text{Fill slice } R_3 \text{ in shard } S_1 \text{ with } N_1 \\ \text{Once } N_1 = 0, \text{ move to the next dataset } N_2 \text{ and repeat the distribution process.} \\ \text{If } N_2 > (n - (N_1 - N_2)), \text{ move to slice } R_3 \text{ of shard } S_2 \text{ and repeat the process until } N_2 = 0. \\ \text{Continue the process for subsequent datasets } N_3, N_4, \ldots \text{ until all data is distributed.} \end{cases}$$

As a result, slice R_3 in shard S_1 will contain data samples from $year_1$ and $year_2$ and others years potentially if ever $N_2 = 0$

This optimized distribution not only prioritizes the most recent and sensitive data—by placing them in the final slices of the earliest shards—but also guarantees balanced allocation across slices, thereby reducing the number of checkpoints that require retraining during deletion requests.

Figure 5 is an example of our selected shard and slice distribution, based on the previously explained logic:

Sample	Year	Recency Score ↑	Sensitivity Score ↑	Data Samples
X_1	2025	$Rr(1)$	$Ss(1)$	140
X_2	2024	$Rr(2)$	$Ss(2)$	90
X_3	2023	$Rr(3)$	$Ss(3)$	100
X_4	2022	$Rr(4)$	$Ss(4)$	170
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
X_N	Oldest Year	$Rr(N)$	$Ss(N)$	N_N

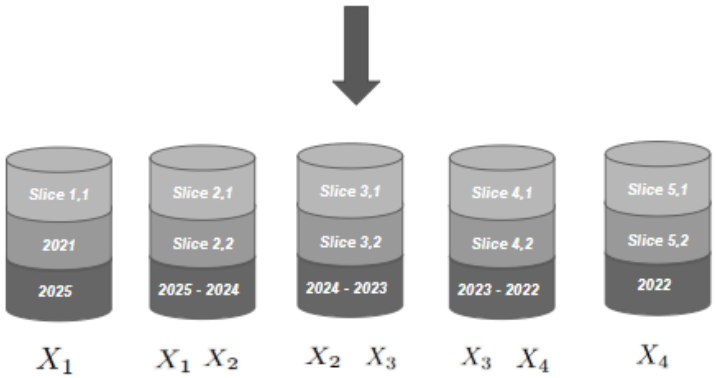

Figure 5: SISA Distribution Strategy Illustration: Following the sharding and slicing logic, we distribute the data into the slices and their corresponding shard

3.3 Training Pipeline

Our training pipeline builds upon the Sharded, Isolated, Sliced, and Aggregated (SISA) framework [17], as presented in Figure 6, with a key enhancement: an optimized data distribution strategy designed to improve unlearning efficiency. Instead of relying on a standard random partitioning approach, we introduce a structured method for distributing data across shards and slices, ensuring that unlearning requests can be processed more effectively.

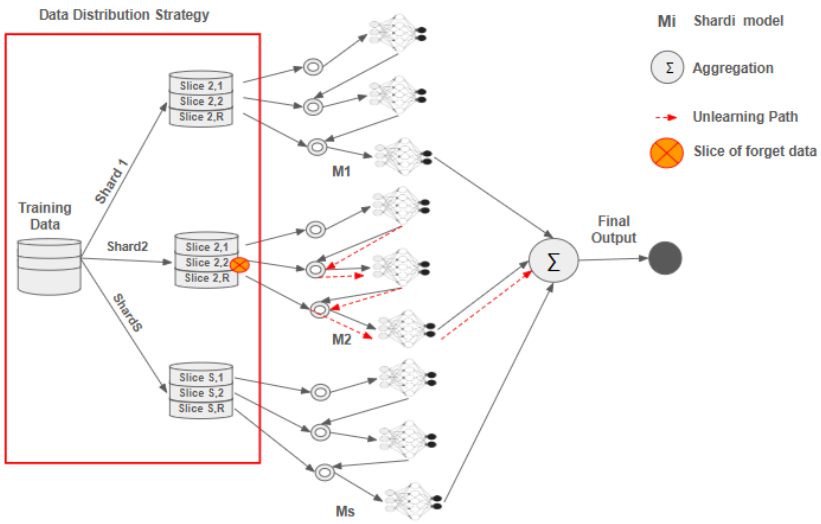

Figure 6: SISA Training Pipeline - The red square highlights our contribution, which specifically targets the data distribution component.

Given a dataset $\mathcal{D}$, we partition it into N_S disjoint *shards* following the logic presented in the previous *Slicing and Sharding* section, ensuring efficient organization for optimized unlearning.

Each shard S_s is further divided into N_R *slices*, represented as:

$$S_s = \bigcup_{r=1}^{N_R} R_{s,r}, \quad \text{where } R_{s,r} \cap R_{s,r'} = \emptyset \text{ for } r \neq r'.$$

This hierarchical structure enables efficient unlearning, as only specific slices within a shard need retraining when an unlearning request is processed. Each slice $R_{s,r}$ undergoes incremental training, meaning that the training process is performed sequentially across slices:

$$\theta_{s,r} = \mathcal{T}(R_{s,r} \mid \theta_{s,r-1}),$$

where $\mathcal{T}$ represents the training function, and $\theta_{s,r}$ is the model state after training on slice $R_{s,r}$. The initial model $\theta_{s,0}$ is typically a pre-trained model.

For inference, we employ an ensemble approach by aggregating predictions across all shards. Given an input x, each shard S_s produces a prediction $\hat{y}_s$, and the final output is determined using *majority voting*:

$$\hat{y} = \arg\max_{y} \sum_{s=1}^{N_S} \mathbb{I}(\hat{y}_s = y),$$

where $\mathbb{I}(\cdot)$ is the indicator function. This aggregation ensures robustness while simplifying the unlearning process [14].

3.4 Unlearning Mechanism

Upon receiving an unlearning request, our system first identifies the relevant shard and slice containing the data to be forgotten. We assume that unlearning requests are not uniformly distributed; instead, they follow an exponential distribution that favors newer data. Specifically, the forgetting probability is modeled as:

$$P_{\text{forget}}(t) \propto e^{-\lambda(t-t_0)},$$

where t represents the publication year, t_0 is a reference year, and λ controls the decay rate, in line with cognitive psychology's forgetting model [28]. For clarity, Table 5 provides an illustrative example showing the forgetting probabilities for various years and their corresponding user request IDs.

Sample	Year	Request IDs	$P_{\textbf{forget}}(t) \uparrow$
X_1	2025	$RQ(1)$	$e^{-\lambda(2025-t_0)}$
X_2	2024	$RQ(2)$	$e^{-\lambda(2024-t_0)}$
X_3	2023	$RQ(3)$	$e^{-\lambda(2023-t_0)}$
X_4	2022	$RQ(4)$	$e^{-\lambda(2022-t_0)}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
X_N	Oldest Year	$RQ(N)$	$e^{-\lambda(\text{Oldest Year}-t_0)}$

Table 5: Forgetting-User Requests Illustration Example

Each unlearning request is processed independently by retraining only the affected slice within its respective shard. After all requests have been individually handled as represented in figure 7, we aggregate the updated model components to form the final model. Specifically, we sum the total unlearning time across all relevant shards and slices, which reflects the overall computational cost of unlearning. Additionally, we calculate the average loss after retraining on the remaining data to measure the impact of unlearning on model performance.

Figure 7: SISA Unlearning Mechanism

This approach ensures efficient unlearning with reduced computational overhead, while also preserving the exact unlearning guarantees and model utility provided by the original SISA framework. By carefully aggregating these updated components—combining both the total unlearning time and the average loss—we maintain a balance between computational efficiency and model performance.

4 Experimental Setup

4.1 Datasets

This study utilizes two distinct datasets, each tailored to specific research objectives:

Copyright Request Dataset The *BookMIA dataset*[2] from Hugging Face is used as the benchmark for evaluating content-based requests, containing 10,000 diverse literary snippets from a wide range of literary works.

Privacy Request Dataset For privacy-focused unlearning tasks, we use the *Wikipedia Biographies Text Generation dataset*[3] from Kaggle, which includes 582,627 training biographies and 72,829 test biographies

extracted from Wikipedia. We only consider a sample of 10,000 from the training set and 2,000 samples from the test set.

4.2 Models

In this study, we employ a BERT-based architecture, specifically `BertForMaskedLM('bert-base -uncased')`, fine-tuned for masked language modeling (MLM) tasks. The original BERT model was pre-trained on two extensive corpora: the **BookCorpus**[4] and a filtered version of **English Wikipedia**[5]. This pre-training involved two primary tasks:

- **Masked Language Modeling (MLM)**: During training, random words in a sentence are replaced with a special `[MASK]` token, and the model learns to predict the original words based on surrounding context.
- **Next Sentence Prediction (NSP)**: Given two sentences, the model learns to predict whether the second sentence follows the first in the original text.

These tasks enable the model to learn deep bidirectional text representations, making it effective for various downstream applications.

4.3 Evaluation Metrics

In most machine unlearning studies, evaluation metrics are primarily designed to verify the correctness and quality of the forgetting process. However, in our case, such verification is unnecessary, as SISA guarantees exact unlearning. Unlike approximate unlearning methods, which do not ensure complete data removal, SISA provides provable guarantees of exact unlearning, as documented in [61]. Consequently, we focus on evaluating the efficiency and utility of the unlearning process using the following key metrics:

- **Unlearning Time**: The duration to complete the unlearning process for specific user forgetting requests, indicating efficiency. This is the time difference between when the unlearning process starts and when it ends.
- **Loss**: Assesses the model's utility, categorized as:
  - *Training Loss*: Measures the error during training and is computed using Cross-Entropy Loss. This loss helps track how well the model fits the training data.
  - *Unlearning Loss*: Measures the training loss after fine-tuning the model on the shard with the forget set removed. It evaluates how well the model adapts to the remaining data once the forget requests have been processed, by measuring its performance during the retraining process.

4.4 Data Distribution and Training Settings

To optimize the unlearning process, the dataset is divided into **5 shards** and **3 slices**, enabling efficient partitioning and targeted unlearning. However, the rationale behind selecting 5 shards and 3 slices could be further explored.

For training, we employ a single epoch with a batch size of **8** for both copyright and privacy-related unlearning requests. Tokenization is performed using the built-in BERT tokenizer, while a modified version with a vocabulary size of **48,755** is used specifically for privacy-related contexts[4].

To enhance representation learning, we integrate `SentenceTransformer('all-MiniLM-L6-v2')`[5], which improves contextual embeddings. Optimization is carried out using the Adam optimizer, balancing performance and computational efficiency.

[2] *BookMIA dataset* refers to a benchmark dataset containing book snippets that are available here: `https://huggingface.co/datasets/swj0419/BookMIA`

[3] *Wikipedia Biographies Text Generation dataset* refers to a dataset containing real biographies and it is available here: https://www.kaggle.com/datasets/thedevastator/wikipedia-biographies-text-generation-dataset/data

[4] *BookCorpus is a dataset comprising approximately 800 million words from over 11,000 unpublished books, commonly used for language model training. See [71] for more details.*

[5] *A structured and curated subset of Wikipedia, containing around 2.5 billion words, used for training large-scale language models. See [37] for further details.*

[4] *To better handle privacy-sensitive data, we modify the BERT tokenizer by incorporating Sentence Transformer to extract full names and add them as distinct tokens. This ensures that the model properly recognizes and processes full names, particularly complex or multi-part names, such as Spanish compound surnames.*

[5] *We used SentenceTransformer('all-MiniLM-L6-v2'), a pre-trained model that generates dense embeddings for text, to extract full names from biographies. It efficiently processes the text to identify and isolate names by capturing the semantic*

4.5 Unlearning Settings

Copyright Requests Our dataset includes 100 books and 10,000 book snippets. For copyright-related unlearning, we choose 10% of the training dataset (equivalent to 10 recently published books that got the highest forgetting probability) as forget requests. The test set for copyright-related unlearning comprises 20 books, totaling 2,000 book snippets.

Privacy Requests For privacy-related unlearning, we utilize a dataset of 10,000 biographies, selecting the 10% with the highest forgetting probability (1,000 biographies) to form the forget set. The test set for privacy-related unlearning consists of 2,000 biographies. Given the impracticality of reporting individual results for each forget request due to the high volume, we only present the top 10 forget requests (the first received) and report the total unlearning time (in seconds) for all requests.

Baseline Comparison To evaluate our unlearning approach, we conduct a comprehensive comparison against four approximate unlearning methods: Gradient Ascent, Gradient Difference, KL Minimization, and Preference Optimization. Additionally, we benchmark our method against three baselines: full retraining, standard SISA, and our enhanced SISA strategy. Finally, we benchmark our SISA and the Standard SISA against both the most probable and randomly selected user requests to evaluate their performance and compare the results.

User Requests Furthermore, for each user request, we implement a sequential unlearning process, as illustrated in Figure 7. Each unlearning request is processed independently, ensuring proper isolation and the final results are aggregated accordingly. For the retraining from scratch baseline, due to its computational cost, we conduct experiments with a single user request. For all other methods, we apply the same unlearning settings for consistency and comparability.

4.6 Baselines

As previously stated, we conduct experiments with various baselines to better assess the impact of our method compared to existing approaches. The baselines considered include:

1. **SISA Enhanced: Working on shards and slices distribution** We plan to begin with the distribution we selected, as described in Section 3.2 - Sharding and Slicing, shown in Figure 8, and then progressively enhance it by incorporating various sensitivity features.

Figure 8: SISA Enhanced: The developed SISA Data Distribution Strategy that incorporates our contribution

The selection of sensitivity features for each type of request is largely driven by the structure and content of the datasets involved. For copyright requests, we consider features such as **Year**, **Textual Similarity**, **Theme**, **Book Length**, and **Semantic Similarity**.

For book snippets, semantic similarity and thematic grouping prove particularly useful. For instance, books published in 2024 may share common themes, and organizing them by theme can accelerate retrieval. This is especially beneficial when user requests come from authors focused on similar themes, a likely scenario.

For instance, authors Mona Awad and Paul Tremblay [5] recently claimed that their books were used without permission by ChatGPT. Both authors wrote books in genres such as dark fantasy and comic fiction, which could be identified more efficiently using a thematic distribution strategy. This approach would also simplify the processing of forgetting requests for books within similar categories.

meaning of each passage.

In the case of privacy requests, the relevant features depend largely on the nature of the biographies and the type of information they contain. Here, we have chosen three key features—**Year**, **Length**, and **Density**—though additional features could be explored. However, their effectiveness might be limited, especially if the users requesting data removal do not share common characteristics like location, Birth Place etc, unless a national privacy regulation is in place.

Ultimately, the choice of features depends on the specifics of the dataset, the context of the requests, and the characteristics of the user requests.

2. **SISA Enhanced: Random Requests** In this experiment, we reproduce the first experiment, but this time the requests are selected randomly. Unlearning requests will be sent to the model without considering the forgetting probability and computations mentioned in Section 3.4 for either type of request.

3. **Approximate Unlearning Implementation:** In this experiment, we explore the approximate unlearning methods outlined in the TOFU paper [20]. The primary objective is to compare our results with those obtained using these approximate methods. While we will use similar evaluation metrics, our main focus will be on unlearning time, as it represents a critical aspect of performance. Notably, when employing these approximate methods, the concepts of shards and slices are no longer considered.

4. **SISA Standard**

   In this experiment, as the name suggests, we apply the standard SISA method, which will serve as our baseline for comparison. Figure 9 represents a sample of how data is normally distributed in Standard SISA.

Figure 9: SISA Standard: The standard SISA Data Distribution

   Specifically, we apply Standard SISA in scenarios involving "more likely to be requested" user requests, as well as in random user requests.

5. **Complete retraining from scratch:** In this experiment, we use the standard retraining-from-scratch method. Figure 10 represent the process of retraining from scratch which includes extracting the forget set and retraining the model on the retain set.

[5] *In 2023, the two authors filed a lawsuit against OpenAI for using their books:* `https://www.theguardian.com/books/2023/jul/05/authors-file-a-lawsuit-against-openai-for-unlawfully-ingesting-their-books`

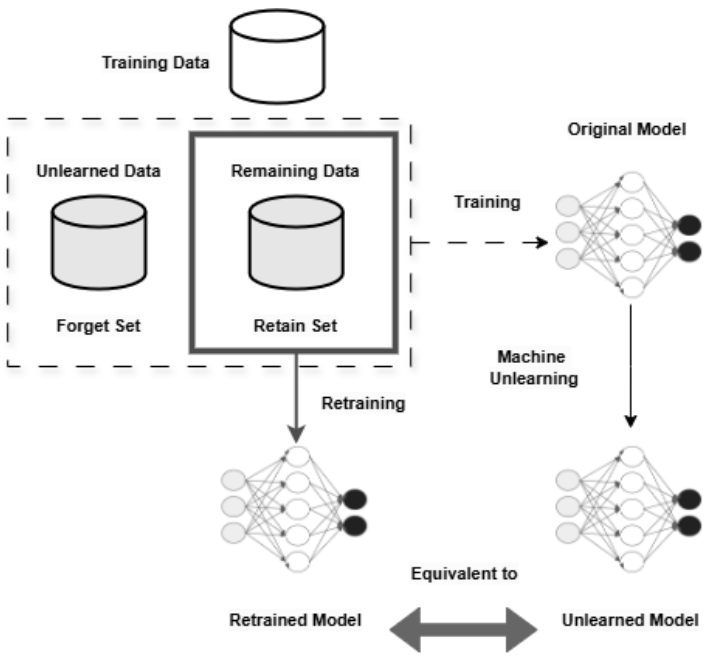

Figure 10: Retraining from Scratch Framework

Retraining from scratch is computationally expensive because each user request to be forgotten is treated as a unique forget set. For every request, the model is retrained on the retain set (i.e., the dataset excluding the forget set). Since this process is repeated independently for each request, it requires retraining the entire model from scratch every time, leading to significant computational costs.

5 Experimental Results

5.1 Copyright Context Requests

In the copyright context, users submitted requests specifying the "Book Name" to have all information related to their books forgotten as highlighted in the figure 10. This includes any extracted content that could lead to a copyright violation.

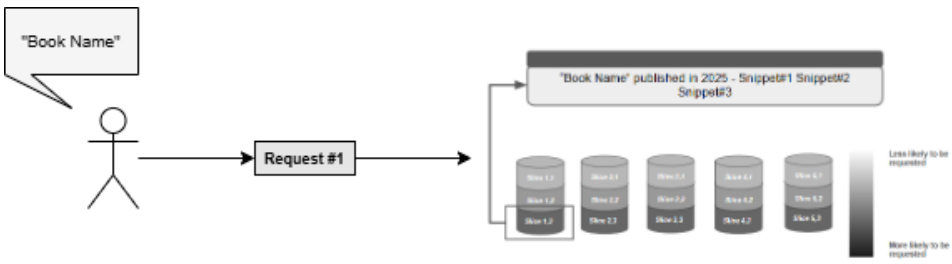

Figure 11: Pipeline for Handling Copyright Requests from the User: The user submits the name of the book to be forgotten, which is treated as a request. Subsequently, the relevant snippets within the corresponding shards and slices are identified.

5.1.1 Enhanced SISA: User Requests (Most likely to be requested)

In Section 3.2 – Forgetting Priority Score, we introduced the sensitivity score, which varies based on the sensitivity features used. As a result, each experiment acquired a different forgetting priority score.

Distribution	Features Used	$Ss(i)$ Formula
1	Publishing Year	0
2	Publishing Year, Textual Similarity	$\gamma_1 Ss_1(i)$
3	Publishing Year, Theme Sensitivity	$\gamma_3 Ss_3(i)$
4	Publishing Year, Snippet Length	$\gamma_2 Ss_2(i)$
5	Publishing Year, Semantic Similarity	$\gamma_4 Ss_4(i)$
6	Publishing Year, Semantic Similarity, Snippet Length	$\gamma_2 Ss_2(i) + \gamma_4 Ss_4(i)$

Table 6: Forgetting Priority Score Formulations Across Different Distributions

(a) **Distribution 1: Using Year as a feature**

(a) Enhanced SISA Training Results

Shard	Slice	Training Loss	Training Time (s)
1	1	0.7214	17.13s
	2	1.4982	
	3	0.8421	
2	1	0.6954	18.00s
	2	0.9321	
	3	0.4823	
3	1	0.3921	18.54s
	2	1.1215	
	3	0.7451	
4	1	0.5812	19.12s
	2	0.6924	
	3	0.2914	
5	1	0.5234	17.13s
	2	0.7435	
	3	0.4123	

(b) Enhanced SISA Unlearning Results

Book Name	Shard	Slice	Unlearning Loss	Unlearning Time (s)
Casino Royale	1	3	0.5436	120.25s
Costanza Casati Clytemnestra	1	3	0.3759	
Treasure Island	2	3	1.5164	
Titanium Noir	2	3	0.4773	
Harry Potter and the Sorcerer's Stone	1	3	0.5600	
Riley Sager The Only One Left	1	3	0.5311	
The Lost Bookshop	2	3	1.4710	
Emma	2	3	0.5093	
Little Women	1	3	0.6085	
The One	3	3	1.7471	

Table 7: SISA Enhanced Results for the first distribution

(b) **Distribution 2: Using Year and Textual Similarity as features**

(a) Shard and Slice Training Results

Shard	Slice	Training Loss	Training Time (s)
1	1	0.6105	18.26s
	2	1.2282	
	3	1.0433	
2	1	1.0161	17.80s
	2	1.1725	
	3	0.9790	
3	1	0.5308	17.48s
	2	1.2218	
	3	0.7612	
4	1	1.2470	19.61s
	2	1.1338	
	3	0.7507	
5	1	0.8267	17.71s
	2	0.8827	
	3	1.4738	

(b) Shard and Slice Unlearning Results

Book Name	Shard	Slice	Unlearning Loss	Unlearning Time (s)
Casino Royale	1	3	1.2785	115.69s
Costanza Casati Clytemnestra	3	3	1.1041	
Treasure Island	1	3	1.3853	
Titanium Noir	1	3	1.5252	
Harry Potter and the Sorcerer's Stone	1	3	1.5040	
Riley Sager The Only One Left	2	3	0.7945	
The Lost Bookshop	1	3	1.0616	
Emma	2	3	0.8093	
Little Women	1	3	1.2070	
The One	2	3	0.8695	

Table 8: SISA Enhanced Results for the second distribution

(c) **Distribution 3: Using Year and Theme as features**

(a) Shard and Slice Training Results

Shard	Slice	Training Loss	Training Time (s)
1	1	0.5634	17.89s
	2	0.8272	
	3	0.9410	
2	1	0.7129	18.56s
	2	0.8836	
	3	0.9471	
3	1	0.6078	19.23s
	2	0.9419	
	3	1.0145	
4	1	1.0674	20.00s
	2	1.1002	
	3	0.8769	
5	1	0.6981	18.23s
	2	0.8130	
	3	1.0982	

(b) Shard and Slice Unlearning Results

Book Name	Shard	Slice	Unlearning Loss	Unlearning Time (s)
Casino Royale	2	3	1.0177	110.73s
Costanza Casati Clytemnestra	2	3	1.0092	
Treasure Island	1	3	1.1454	
Titanium Noir	1	3	0.8658	
Harry Potter and the Sorcerer's Stone	1	3	0.9819	
Riley Sager The Only One Left	1	3	0.9364	
The Lost Bookshop	1	3	1.0634	
Emma	1	3	1.1398	
Little Women	2	3	0.6013	
The One	2	3	1.6480	

Table 9: SISA Enhanced Results for the third distribution

(d) **Distribution 4: Using Year and Length as features**

(a) Shard and Slice Training Results

Shard	Slice	Training Loss	Training Time (s)
1	1	0.6845	19.20s
	2	0.8976	
	3	0.9781	
2	1	0.7342	18.74s
	2	0.8394	
	3	1.0120	
3	1	0.5679	19.60s
	2	0.6654	
	3	0.9441	
4	1	0.7218	18.93s
	2	0.8119	
	3	0.9702	
5	1	0.8432	19.11s
	2	0.9125	
	3	1.0293	

(b) Shard and Slice Unlearning Results

Book Name	Shard	Slice	Unlearning Loss	Unlearning Time (s)
Casino Royale	1	3	1.7130	109.18s
Costanza Casati Clytemnestra	1	3	0.9738	
Treasure Island	1	3	1.9490	
Titanium Noir	1	3	0.9888	
Harry Potter and the Sorcerer's Stone	1	3	0.5965	
Riley Sager The Only One Left	1	3	0.9729	
The Lost Bookshop	2	3	1.3692	
Emma	2	3	0.3519	
Little Women	3	3	0.7290	
The One	2	3	1.3468	

Table 10: SISA Enhanced Results for the fourth distribution

(e) **Distribution 5: Using Year and Semantic Similarity as features**

(a) Enhanced SISA Training Results

Shard	Slice	Training Loss	Training Time (s)
1	1	0.6327	18.23s
	2	0.7349	
	3	0.9215	
2	1	0.7463	19.12s
	2	0.8132	
	3	1.0391	
3	1	0.8221	17.98s
	2	0.6898	
	3	1.0805	
4	1	0.9845	18.49s
	2	0.9220	
	3	0.7467	
5	1	0.8021	17.74s
	2	0.9516	
	3	1.0468	

(b) Shard and Slice Unlearning Results

Book Name	Shard	Slice	Unlearning Loss	Unlearning Time (s)
Casino Royale	1	3	0.9476	107.97s
Costanza Casati Clytemnestra	1	3	0.8926	
Treasure Island	1	3	0.8829	
Titanium Noir	1	3	1.2068	
Harry Potter and the Sorcerer's Stone	1	3	1.2049	
Riley Sager The Only One Left	1	3	0.9816	
The Lost Bookshop	2	3	1.4189	
Emma	2	3	0.4306	
Little Women	3	3	0.3418	
The One	1	3	1.9671	

Table 11: SISA Enhanced Results for the fifth distribution

(f) **distribution 6: Using Year, Semantic Similarity and Length as features**

(a) Enhanced SISA Training Results

Shard	Slice	Training Loss	Training Time (s)
1	1	0.7354	18.03s
	2	0.8611	
	3	0.9265	
2	1	0.6123	17.82s
	2	0.7012	
	3	0.9240	
3	1	0.8241	19.44s
	2	0.9443	
	3	1.0276	
4	1	0.6128	18.95s
	2	0.7824	
	3	0.9132	
5	1	0.6712	18.17s
	2	0.7485	
	3	0.8532	

(b) Shard and Slice Unlearning Results

Book Name	Shard	Slice	Unlearning Loss	Unlearning Time (s)
Casino Royale	1	3	0.6956	96.27s
Costanza Casati Clytemnestra	1	3	1.0190	
Treasure Island	1	3	1.9777	
Titanium Noir	1	3	0.9670	
Harry Potter and the Sorcerer's Stone	1	3	0.5818	
Riley Sager The Only One Left	1	3	0.9694	
The Lost Bookshop	2	3	1.3679	
Emma	1	3	0.3556	
Little Women	2	3	0.7208	
The One	1	3	1.3435	

Table 12: SISA Enhanced Results for the sixth distribution

5.1.2 Approximate methods

Approximate Method	Training Time (s)	Unlearning Time (s)	Training Loss	Unlearning Loss
Gradient Ascent	339.36s	102.80s	2.0510	1.2337
Gradient Difference		103.20s		1.3422
KL Minimization		101.00s		2.3937
Preference Optimization		105.30s		2.0548

Table 13: Approximate Methods Results: Copyright Requests

5.1.3 Enhanced SISA: Random user requests

The objective behind this experiment as stated before is to actually get a general idea of how SISA behaves when the requests are random, so we are showcasing the first three results of our experiments.

(a) **Distribution 1: Using Year as a feature**

Table 14: Enhanced SISA Unlearning for the first distribution: Random user requests

Book Name	Shard	Slice	Unlearning Loss	Unlearning Time (s)
Things Fall Apart	1	1	0.9821	242.52s
		2	0.7453	
		3	0.8519	
We Could Be So Good	3	3	1.0324	
		3	1.0896	
Dracula	4	2	1.5023	
		3	1.3204	
The Invisible Man	3	2	0.7432	
		3	0.8125	
Alex Hay The Housekeeper	2	2	1.2715	
		3	0.6954	
Happy Place	5	3	1.2093	
The Mysteries of Udolpho	4	2	0.8912	
		3	1.0658	
A Day Of Fallen Night	3	1	0.5612	
		2	0.8924	
		3	0.8039	
Romantic Comedy	5	1	1.5029	
		2	0.9256	
		3	0.7048	
The One	2	3	0.8976	

(b) **Distribution 2: Using Year and Textual Similarity as features**

Table 15: Enhanced SISA Unlearning for the second distribution: Random user requests

Book Name	Shard	Slice	Unlearning Loss	Unlearning Time (s)
Things Fall Apart	1	1	1.0124	
		2	0.7542	
		3	0.8349	
We Could Be So Good	4	1	1.0293	
		2	0.9927	
		3	1.0824	
Dracula	1	2	1.4328	
		3	1.2095	
The Invisible Man	2	1	0.7126	
		2	0.8032	
		3	0.8569	
Alex Hay The Housekeeper	3	3	1.3215	230.58s
Happy Place	2	2	1.0675	
		3	1.0432	
The Mysteries of Udolpho	5	2	0.8936	
		3	1.0241	
A Day Of Fallen Night	3	2	0.9418	
		3	0.7893	
Romantic Comedy	5	2	0.9241	
		3	0.6928	
The One	2	3	0.9031	

(c) **Distribution 3: Using Year and Theme as features**

Table 16: Enhanced SISA Unlearning for the third distribution: Random user requests

Book Name	Shard	Slice	Unlearning Loss	Unlearning Time (s)
Things Fall Apart	5	1	0.9874	
		2	0.8235	
		3	0.7593	
We Could Be So Good	4	1	1.0234	
		2	1.0156	
		3	0.9962	
Dracula	1	1	1.5432	
		2	1.3487	
		3	1.1267	
The Invisible Man	1	1	0.6732	
		2	0.8913	
		3	0.8291	
Alex Hay The Housekeeper	3	3	1.4123	222.68s
Happy Place	2	2	1.1342	
		3	0.9832	
The Mysteries of Udolpho	5	2	0.9266	
		3	1.0412	
A Day Of Fallen Night	3	2	0.9315	
		3	0.8013	
Romantic Comedy	1	2	0.8467	
		3	0.7564	
The One	3	3	0.9324	

5.1.4 SISA Standard

(a) **SISA Standard: User Requests (Most likely to be requested)**

(a) Shard and Slice Training Results

Shard	Slice	Training Loss	Training Time (s)
1	1	0.6312	18.83s
	2	0.7723	
	3	0.9260	
2	1	0.5597	18.10s
	2	0.7831	
	3	0.8602	
3	1	0.6887	19.40s
	2	0.7674	
	3	0.9832	
4	1	0.7201	18.21s
	2	0.8855	
	3	0.9120	
5	1	0.7922	18.65s
	2	0.8912	
	3	0.9863	

(b) Shard and Slice Unlearning Results

Book Name	Shard	Slice	Unlearning Loss	Unlearning Time (s)
Casino Royale	1	1	1.0620	280.37s
		2	0.8293	
		3	0.8674	
Costanza Casati Clytemnestra	1	3	0.9956	
Treasure Island	5	1	1.6895	
		2	1.1718	
		3	1.0559	
Titanium Noir	4	1	0.6656	
		2	0.9090	
		3	0.8355	
Harry Potter and the Sorcerer's Stone	2	2	1.4176	
		3	0.7125	
Riley Sager The Only One Left	5	3	1.1000	
The Lost Bookshop	5	2	0.9266	
		3	1.0438	
Emma	4	1	0.5419	
		2	0.9396	
		3	0.8101	
Little Women	3	1	1.6052	
		2	0.8624	
		3	0.6673	
The One	2	3	0.9671	

Table 17: SISA Standard Unlearning Results - User Requests

(b) **SISA Standard: Random user requests**

Table 18: SISA Standard Unlearning Results - Random Requests

Book Name	Shard	Slice	Unlearning Loss	Unlearning Time (s)
Things Fall Apart	5	1	1.0321	305.45s
		2	0.8429	
		3	0.7983	
We Could Be So Good	1	1	0.9124	
		2	0.8793	
		3	0.8567	
Dracula	3	2	1.5321	
		3	1.4723	
The Invisible Man	5	3	1.1489	
		2	0.8753	
		3	0.8327	
Alex Hay The Housekeeper	5	2	1.3657	
		3	1.2984	
Happy Place	4	1	1.1053	
		2	1.0832	
		3	1.0345	
The Mysteries of Udolpho	2	2	0.9536	
		3	1.0278	
A Day Of Fallen Night	2	1	0.9154	
		2	0.8872	
		3	0.7983	
Romantic Comedy	1	2	0.8321	
		3	0.7632	
The One	2	3	0.9531	

5.1.5 Retraining From scratch: Case of one user request

Retraining from scratch serves as one of the baseline experiments against which we compare our Enhanced SISA approach. Below, we present the results of retraining from scratch for a single user request. In this experiment, each request is handled independently, rather than processing all requests simultaneously.

Based on this approach, we estimate that retraining the model from scratch for each of the 10 user requests would result in a total time of approximately 3370 seconds which is roughly 1 Hour.

Method	Training Time (s)	Training Loss	Unlearning Time (s)	Unlearning Loss
Retraining From Scratch	339.36s	2.0510	337.61s	2.0620

Table 19: Performance metrics for the retraining from scratch method.

5.2 Privacy Context Requests

In the privacy context, users submitted requests using their "Full Name," including those composed of three different parts, to ensure the removal of all information specifically related to them, as illustrated in Figure 12. This also includes any mentions of their name in other biographies.

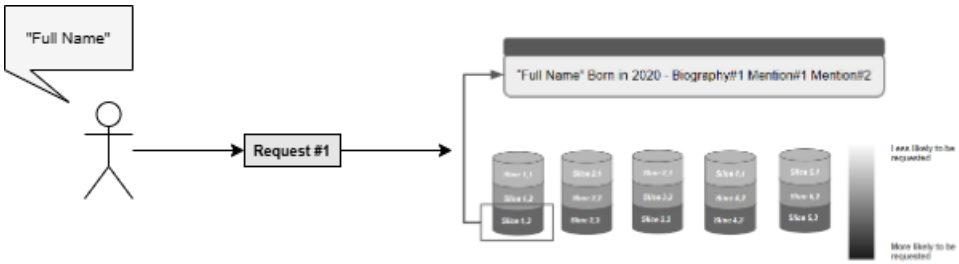

Figure 12: Pipeline of the Privacy Request Received by the User

5.2.1 Enhanced SISA: User Requests (Most likely to be requested)

In the privacy context, we utilize two distinct sensitivity features. After testing, incorporating additional features such as Location, Job, or Hobbies proved impractical due to the diverse and varied nature of the biographies. However, if a law were legislated in a specific country, Location could serve as a relevant sensitivity metric, as it would likely trigger a significant number of requests for data removal from that region.

Distribution	Features Used	$Ss(i)$ Formula
1	Birth Year	0
2	Birth Year, Length	$\gamma_1 Ss_1(i)$
3	Birth Year, Density	$\gamma_2 Ss_2(i)$

Table 20: Forgetting Priority Score Formulations Across Different Distributions

(a) **Distribution 1: Using Year as a feature**

Table 21: Enhanced SISA Results for the first distribution

(a) Enhanced SISA Training

Shard	Slice	Training Loss	Training Time (s)
1	1	1.6265	44.05s
	2	1.5136	
	3	1.7199	
2	1	1.6278	43.28s
	2	1.4158	
	3	1.3695	
3	1	1.6400	43.44s
	2	1.4106	
	3	1.3396	
4	1	1.5732	43.82s
	2	1.4457	
	3	1.3471	
5	1	1.6580	43.70s
	2	1.5278	
	3	1.4182	

(b) Enhanced SISA Unlearning

User's Name	Shard	Slice	Unlearning Loss	Unlearning Time (s)	Total Time (s)
denis louis blackmore	1	3	2.2876	139.67s	23,877.09s
sohaila zaland	1	3	2.0983		
kelli lidell	1	3	2.3754		
stephen a. rotter	1	3	2.5127		
thomas tom whittaker	1	3	2.4932		
susan batson	1	3	1.8012		
kramer morgenthau	1	3	2.0723		
ali lotfi	1	3	1.8174		
eloïne barry	1	3	2.2135		
saint leucius	1	3	1.8618		

(b) **Distribution 2: Using Year and Length as features**

(a) Enhanced SISA Training

Shard	Slice	Training Loss	Training Time (s)
1	1	1.5828	43.04s
	2	1.4869	
	3	1.7081	
2	1	1.6157	42.99s
	2	1.4462	
	3	1.3258	
3	1	1.6245	42.14s
	2	1.4384	
	3	1.3417	
4	1	1.5732	42.28s
	2	1.4457	
	3	1.3471	
5	1	1.6580	42.53s
	2	1.5278	
	3	1.4182	

(b) Enhanced SISA Unlearning

User's Name	Shard	Slice	Unlearning Loss	Unlearning Time (s)	Total Time (s)
denis louis blackmore	1	3	2.2458	139.13s	23,054.93s
sohaila zaland	1	3	2.1234		
kelli lidell	1	3	2.4017		
stephen a. rotter	1	3	2.5089		
thomas tom whittaker	1	3	2.4896		
susan batson	1	3	1.7823		
kramer morgenthau	1	3	2.0754		
ali lotfi	1	3	1.8246		
eloïne barry	1	3	2.1935		
saint leucius	1	3	1.8549		

Table 22: Enhanced SISA Results for the second distribution

(c) **Distribution 3: Using Year and Density as sensitivity features**

(a) Enhanced SISA Training

Shard	Slice	Training Loss	Training Time (s)
1	1	1.6115	43.19s
	2	1.5116	
	3	1.7310	
2	1	1.6073	43.23s
	2	1.4540	
	3	1.3880	
3	1	1.6536	43.35s
	2	1.4572	
	3	1.3779	
4	1	1.5761	43.30s
	2	1.4347	
	3	1.4016	
5	1	1.6377	43.70s
	2	1.4675	
	3	1.4217	

(b) Enhanced SISA Unlearning

User's Name	Shard	Slice	Unlearning Loss	Unlearning Time (s)	Total Time (s)
denis louis blackmore	1	3	2.2634	138.73	22,946.12s
sohaila zaland	1	3	2.1189		
kelli lidell	1	3	2.3721		
stephen a. rotter	1	3	2.5124		
thomas tom whittaker	1	3	2.4978		
susan batson	1	3	1.8074		
kramer morgenthau	1	3	2.0793		
ali lotfi	1	3	1.8176		
eloïne barry	1	3	2.1910		
saint leucius	1	3	1.8593		

Table 23: Enhanced SISA Results for the third distribution

5.2.2 Approximate methods

Approximate Method	Training Time (s)	Unlearning Time (s)	Training Loss	Unlearning Loss
Gradient Ascent	1340.51s	10,792.91s	1.6852	2.3214
Gradient Difference		13,238.54s		3.6678
KL Minimization		11,303.10s		2.5209
Preference Optimization		12,485.27s		2.3342

Table 24: Approximate Methods Results: Privacy Requests

5.2.3 Enhanced SISA: Random user requests

(a) **Distribution 1: Using Year as a feature**

Table 25: Enhanced SISA Unlearning for the first distribution: Random user requests

User's Name	Shard	Slice	Unlearning Loss	Unlearning Time (s)	Total Time (s)
leonidas alcidas giroux	5	2	3.1111	354,25s	39,120.10s
		3	2.7745		
john brabender	1	2	2.8257		
		3	1.5622		
fahd azam	1	1	3.2926		
		2	2.3489		
		3	2.1938		
tom brower	1	1	2.9059		
		2	2.8972		
		3	3.0421		
josé eloy palacios	1	1	2.8183		
		2	2.1481		
		3	2.4354		
rené froger	5	2	2.4713		
		3	2.1366		
jessica eye	5	3	2.1416		
wallace de souza	4	1	2.1562		
		2	2.0710		
		3	2.1004		
alick jeffrey	5	2	2.1405		
		3	2.0562		
james k irving	3	1	2.0995		
		2	2.1384		
		3	2.2263		

(b) **Distribution 2: Using Year and Length as features**

Table 26: Enhanced SISA Unlearning for the second distribution: Random user requests

User's Name	Shard	Slice	Unlearning Loss	Unlearning Time (s)	Total Time (s)
leonidas alcidas giroux	5	2	2.6637		
		3	1.3308		
john brabender	1	2	2.4691		
		3	1.6565		
fahd azam	1	1	3.7188		
		2	2.6335		
		3	2.3820		
tom brower	1	1	1.1207		
		2	2.4066		
		3	2.2082		
josé eloy palacios	5	3	2.5655	321,17s	38,740.09s
rené froger	5	3	2.3932		
jessica eye	4	1	1.5106		
		2	1.1275		
		3	2.1903		
wallace de souza	5	2	2.1765		
		3	1.0751		
alick jeffrey	3	1	2.1375		
		2	2.1750		
		3	1.0609		
james k irving	3	1	2.1291		
		2	1.1727		
		3	2.3165		

(c) **Distribution 3: Using Year and Semantic Similarity as features**

Table 27: Enhanced SISA Unlearning for the third distribution: Random user requests

User's Name	Shard	Slice	Unlearning Loss	Unlearning Time (s)	Total Time (s)
leonidas alcidas giroux	5	2	2.0412		
		3	1.3295		
john brabender	1	2	2.1504		
		3	1.8708		
fahd azam	1	1	2.7439		
		2	2.1110		
		3	2.0571		
tom brower	1	1	1.4123		
		2	2.2481		
		3	2.1920		
josé eloy palacios	5	3	2.3187	318.10s	38,661.22s
rené froger	4	3	2.1985		
jessica eye	4	1	1.6082		
		2	1.1003		
		3	2.0847		
wallace de souza	5	2	2.1136		
		3	1.0654		
alick jeffrey	3	1	2.0705		
		2	2.1507		
		3	1.0225		
james k irving	3	1	2.0217		
		2	1.1214		
		3	2.1203		

5.2.4 SISA Standard

(a) **SISA Standard: User Requests (Most likely to be requested)**

Table 28: SISA Standard Results

(a) SISA Standard Training

Shard	Slice	Training Loss	Training Time (s)
1	1	1.6047	43.55s
	2	1.4945	
	3	1.3824	
2	1	1.6262	42.61s
	2	1.4411	
	3	1.4397	
3	1	1.6275	42.08s
	2	1.5147	
	3	1.4457	
4	1	1.6418	43.08s
	2	1.4946	
	3	1.5126	
5	1	1.6012	42.12s
	2	1.5297	
	3	1.5124	

(b) SISA Standard Unlearning

User's Name	Shard	Slice	Unlearning Loss	Unlearning Time (s)	Total Time (s)
denis louis blackmore	4	1	3.1907		
		2	3.0661		
		3	2.5272		
sohaila zaland	3	3	2.3685		
kelli lidell	1	1	2.3685		
stephen a. rotter	2	2	3.1395		
		3	2.2224		
thomas tom whittaker	1	1	2.2644	477.05s	40,550.40
		2	2.3227		
		3	3.2123		
susan batson	5	1	3.3413		
		2	3.1773		
		3	2.8173		
kramer morgenthau	5	1	2.4736		
		2	2.1786		
		3	2.8173		
ali lotfi	4	1	2.2224		
		2	3.0947		
		3	2.1240		
eloïne barry	2	1	2.2309		
		2	2.3156		
		3	2.1297		
saint leucius	2	1	3.0661		
		2	2.3717		
		3	2.8635		

(b) **SISA Standard: Random user requests**

Table 29: SISA Standard Unlearning: Random Requests

User's Name	Shard	Slice	Unlearning Loss	Unlearning Time (s)	Total Time (s)
leonidas alcidas giroux	4	1	2.1845		
		2	2.2378		
		3	2.3992		
john brabender	4	1	2.8954		
		2	3.1823		
		3	2.8397		
fahd azam	5	3	3.1684		
tom brower	5	1	2.3298		
		2	2.5731		
		3	3.3817		
josé eloy palacios	3	1	2.3489	364,12s	41,320.45
		2	2.5924		
		3	2.3768		
rené froger	5	2	2.4973		
		3	2.1462		
jessica eye	5	3	2.1584		
wallace de souza	4	1	2.1736		
		2	2.0953		
		3	2.1187		
alick jeffrey	5	2	2.1612		
		3	2.0714		
james k irving	1	1	2.1183		
		2	2.1537		
		3	2.2428		

5.2.5 Retraining From scratch: Case of one User Request

Below, we present the results of retraining from scratch for a single user request. We estimate that retraining the model from scratch for each of the 1000 user requests would result in a total time of approximately 163,9170 seconds which is roughly 455.33 hours.

Method	Unlearning Time (s)	Training Loss	Unlearning Time (s)	Unlearning Loss
Retraining From Scratch	1340.51	1.68520	1639.17	1.85191

Table 30: Performance metrics for the retraining from scratch method.

6 Discussions and Conclusion

Discussions Next, we will discuss the results of each experiment conducted, examining the outcomes separately for both the copyright and privacy contexts.

(a) **Copyright Context Requests**

Table 31: Copyright Requests - Final Evaluation

Experiments	Training		Unlearning	
	Training Time (s)	Training Loss	Unlearning Time (s) ↓	Unlearning Loss
Sixth Distribution	92.41s	0.8092	96.27s	1.0997
KL Minimization	339.36s	2.0510	101.00s	2.3937
Gradient Ascent	339.36s	2.0510	102.80s	1.2337
Gradient Difference	339.36s	2.0510	103.20s	1.3422
Preference Optimization	339.36s	2.0510	105.30s	2.0548
Fourth Distribution (SISA Enhanced)	95.58s	0.8408	109.18s	1.0991
Third Distribution (SISA Enhanced)	93.91s	0.8729	110.73s	1.0708
Second Distribution (SISA Enhanced)	90.86s	0.9918	115.69s	1.0545
First Distribution (SISA Enhanced)	89.92s	0.7116	120.25s	1.5381
SISA Standard	93.19s	0.8184	280.37s	0.9941
Third Distribution (SISA Enhanced Random Requests)	93.91s	0.8729	222.68s	1.0104
Second Distribution (SISA Enhanced Random Requests)	90.86s	0.9918	230.58s	1.0270
First Distribution (SISA Enhanced Random Requests)	89.92s	0.7116	242.52s	0.9975
SISA Standard Random Requests	93.19s	0.8184	305.45s	1.0084
Retraining on Retain Set	339.36s	2.0510	337.61s	2.0620

The results in Table 31 provide an insightful comparison of various methods for training and unlearning in copyright-related requests. The **SISA Standard** methods demonstrate high unlearning times (280.37s–305.45s) with consistent training losses (0.8184) and relatively low unlearning losses (0.9941–1.0084). However, the training time (93.19s) for SISA Standard is comparable to SISA Enhanced, highlighting limited efficiency improvements.

In contrast, **SISA Enhanced** methods achieve a noticeable reduction in unlearning times, particularly for the Sixth Distribution, which demonstrates the lowest unlearning time (96.27s) while maintaining a reasonable unlearning loss (1.0997). Among the enhanced distributions, the Fifth Distribution achieves the best balance, with a low unlearning loss (1.0275) and moderate unlearning time (107.97s), making it a strong candidate for an efficient, accurate and exact unlearning method.

Approximate Unlearning methods provide quick unlearning times (101.00s–105.30s), but they incur the highest training losses (2.0510) and significant unlearning losses (1.2337–2.3937), especially for KL Minimization (2.3937). This illustrates the trade-offs between efficiency and accuracy in heuristic-based approaches.

The experiments involving **SISA Enhanced Random Requests** demonstrate longer unlearning times compared to the standard SISA Enhanced experiments; for example, the unlearning time for the First Distribution is (242.52s). This outcome is expected due to the use of random user requests. Despite the increased unlearning times, the unlearning losses range from (0.9975) to (1.0270) and are comparable to those seen in SISA Enhanced experiments. Furthermore, both SISA Enhanced and SISA Standard Random Requests outperform SISA Standard in terms of efficiency and accuracy.

This indicates that while **SISA Enhanced** does provide some improvement over **SISA Standard** when handling random user requests, both methods encounter difficulties in this context. The randomness of user requests leads to inefficiencies within both SISA configurations, as evidenced by the increased unlearning times. Overall, SISA, whether in its enhanced or standard form, does not perform optimally under random user request conditions, although the enhanced version does manage to surpass the standard variant.

Retraining on the Retain Set yields the longest unlearning time (337.61s) and the highest unlearning loss (2.0620), making it the least efficient approach. It is important to note that this approach is based on retraining from scratch for a single request, which results in a high unlearning time. When applied to multiple requests, such as 10 requests, this method would achieve the highest unlearning time (approximately 3370.61s), further emphasizing its inefficiency in terms of computational resources.

Overall, while SISA Enhanced provides significant improvements over the standard SISA and approximate methods, its performance varies across distributions, with the Fifth and Sixth Distributions emerging as the most balanced configurations for unlearning tasks.

(b) **Privacy Context Requests**

Table 32: Privacy Requests - Final Evaluation

Experiments	Training		Unlearning	
	Training Time (s)	Training Loss	Unlearning Time (s) ↓	Unlearning Loss
Gradient Ascent (Approximate Unlearning)	1340.51s	1.6852	10,792.91s	2.3214
KL Minimization (Approximate Unlearning)	1340.51s	1.6852	11,303.10s	2.5209
Preference Optimization (Approximate Unlearning)	1340.51s	1.6852	12,485.27s	2.3342
Gradient Difference (Approximate Unlearning)	1340.51s	1.6852	13,238.54s	3.6678
Third Distribution (SISA Enhanced)	421.00s	1.5154	22,946.12s	2.0673
Second Distribution (SISA Enhanced)	414.01s	1.5027	23,054.93s	2.1500
First Distribution (SISA Enhanced)	420.93s	1.5089	23,877.09s	2.1533
Third Distribution (SISA Enhanced Random Requests)	421.00s	1.5154	38,661.22s	2.0917
Second Distribution (SISA Enhanced Random Requests)	414.01s	1.5027	38,740.09s	2.0922
First Distribution (SISA Enhanced Random Requests)	420.93s	1.5089	39,120.10s	2.5034
SISA Standard	413.69s	1.5246	40,550.40s	2.1519
SISA Standard Random Requests	413.69s	1.5246	41,320.45s	2.4112
Retraining on Retain Set	1340.51s	1.6852	1,639.17s	1.8519

The results in Table 32 offer a comprehensive evaluation of various methods for training and unlearning in privacy-related requests. The **SISA Standard** methods demonstrate extremely high unlearning times (40550.40s–41320.45s), which is much greater compared to the copyright-related requests, likely due to the larger dataset size (900 privacy requests) used in these experiments. Despite the high unlearning times, the training losses remain consistent (1.5246), and unlearning

losses are relatively high (2.1519–2.4112). These results highlight that SISA Standard is highly inefficient when handling privacy requests, especially with a larger dataset.

On the other hand, the **SISA Enhanced** methods results in high unlearning times, ranging from 22946.12 seconds to 23877.09 seconds. However, they outperform the Standard SISA experiments in both training and unlearning times. Despite this improved efficiency, the unlearning losses, which range from 2.0673 to 2.1533, remain significant. This indicates that, even with advancements over the standard method, SISA Enhanced still faces challenges in handling privacy requests, particularly when dealing with large datasets, such as 10000 biographies.

The **Approximate Unlearning** methods typically provide faster unlearning times (ranging from 10792.91 to 13238.54 seconds). However, they come with significant trade-offs in accuracy, resulting in high unlearning losses (between 2.3214 and 3.6678). These findings highlight that although approximate methods are more efficient, they tend to be less accurate, particularly for complex requests related to privacy, where precision is essential and critical.

The **SISA Enhanced Random Requests** experiments show longer unlearning times compared to the SISA Enhanced experiments, with the First Distribution showing the longest unlearning time (39120.10s). This increase is expected due to the randomness of user requests, which complicates the unlearning process. Despite the increased unlearning times, the unlearning losses (2.0917–2.5034) remain within a similar range to those observed in SISA Enhanced, but with some variations.

Both the SISA Enhanced Random Requests and SISA Standard Random Requests outperform the SISA Standard in terms of unlearning efficiency and accuracy. However, they still face challenges when handling random user requests. This suggests that while the **SISA Enhanced** option does offer some improvements over the **SISA Standard** for privacy requests, both configurations struggle with inefficiencies when user requests are random. The unpredictability of these random requests leads to significantly longer unlearning times.

Retraining on the Retain Set results in the longest unlearning times, with one request taking approximately (1,639.17) seconds. This inefficiency becomes even more evident when applied to multiple requests. For instance, if applied to 900 requests, this method would incur a total unlearning time of around (163,9170) seconds, highlighting its significant computational inefficiency.

In summary, while **SISA Enhanced** offers improvements over **SISA Standard** and approximate methods in some experiments—particularly when receiving more structured privacy requests—it struggles with the complexity and size of larger datasets. The performance is significantly impacted by larger datasets, and random requests only make these challenges worse. This highlights the necessity for further optimization of SISA frameworks to effectively manage larger and more complex datasets.

Conclusion In this work, we revisited the SISA framework with the goal of improving its data distribution strategy to enhance the efficiency of unlearning. The standard SISA approach introduces randomness in data distribution, which can lead to longer unlearning times, especially as the volume of user requests increases due to growing awareness of privacy rights and copyright concerns. Our objective was to optimize SISA by structuring the dataset more effectively. This restructuring aims to achieve both precise and efficient unlearning—precision is ensured by the SISA framework, while efficiency is achieved through an organized data distribution strategy using our priority forgetting score.

Despite these improvements, our approach still has limitations, particularly when dealing with random user requests. While SISA Enhanced performs better than Standard SISA, it still exhibits inefficiencies. Moreover, privacy-related requests present greater challenges due to the complexity and scale of the dataset, resulting in longer unlearning times compared to copyright-related requests.

One important aspect that remains unexplored is the choice of the number of shards and slices. While our experiments assumed an optimal configuration, we did not formally justify this choice. A crucial area for future research is to systematically explore how different shard and slice configurations affect unlearning efficiency and accuracy. This ensures that the chosen partitioning strategy effectively reduces computational overhead while maintaining model performance.

Our findings highlight the potential of structured data distribution in improving SISA, while also highlighting areas that need further refinement, particularly in optimizing SISA for large-scale and dynamic unlearning contexts.

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A Data Distribution Strategies

Before adopting the strategy outlined in Section 3.2 - Sharding and Slicing, we considered an alternative distribution strategy that we believed would be more efficient. This alternative focused on slices rather than shards.

The approach involved initially gathering all the highest priority score elements within the slices of the first shard. Once this shard reached its maximum capacity, we would then move on to the next shard. Figure 13 represents the data distribution strategy with the *more likely to be requested* placed in the first shard and *less likely to be requested* placed in the latest shard.

Figure 13: Data Distribution Strategy Following the Slices

The distribution strategy did not perform optimally because some recent years, considered the most likely requests, were placed in slice 1,1. This arrangement is impractical as it requires retraining all the slices associated with shard 1.

Therefore, we thought it could be better to focus on shards rather than slices, offering then the opportunity for samples with the highest priority score to be placed initially in the third slice of each shard, and then we move forward following the capacity of each slice. Figure 14 illustrates our selected strategy, which delivered better results by placing the *more likely to be requested* in the latest slices of each shard, thereby reducing the need for slice retraining.

Figure 14: Data Distribution Strategy Following the Shards

Furthermore, when handling multiple user forget requests, we often ended up clearing the entire shard as represented in the figure 15, leading to significant data loss. This unintended consequence negatively impacted the model's utility, which was contrary to our objectives.

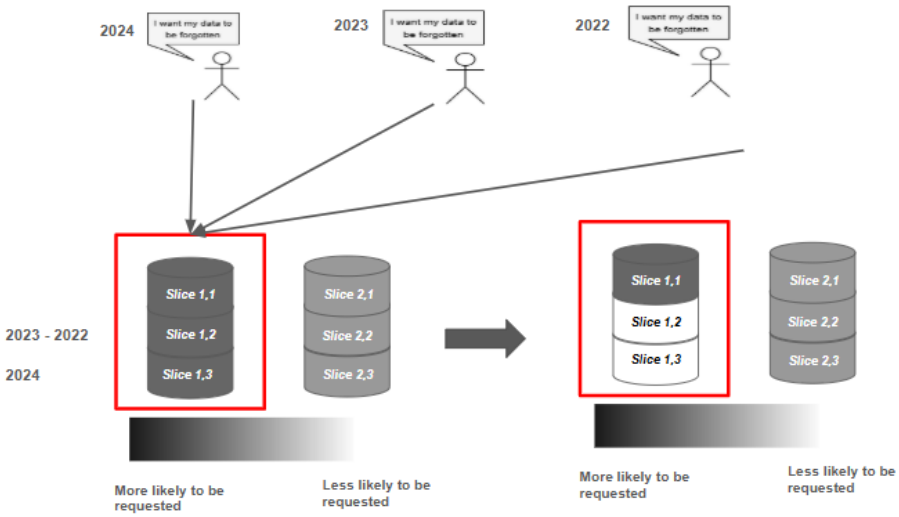

Figure 15: Data distribution strategy based on the slices receiving three forget user requests, leading to the clearing of the second and third slices in the first shard.

B Sensitivity Features Explanation

Throughout our experiments, we computed various sensitivity features to refine and optimize our data distribution strategies. Each feature was derived based on its underlying logic, making it essential to explain its significance and mathematical formulation. The key sensitivity features we considered include:

Text Similarity Text similarity measures lexical overlap between two texts, helping detect redundant content. We used TF-IDF vectorization and computed the cosine similarity as:

$$\text{TextSim}(T_1, T_2) = \frac{\sum_{i=1}^{n} T_1^i T_2^i}{\sqrt{\sum_{i=1}^{n} (T_1^i)^2} \cdot \sqrt{\sum_{i=1}^{n} (T_2^i)^2}} \tag{13}$$

where T_1 and T_2 are TF-IDF vector representations of the text samples. Higher values indicate greater lexical similarity.

Semantic Similarity Semantic similarity captures meaning rather than exact word matches. We used BERT embeddings and calculated cosine similarity between sentence embeddings:

$$\text{SemSim}(S_1, S_2) = \frac{\mathbf{E}(S_1) \cdot \mathbf{E}(S_2)}{\|\mathbf{E}(S_1)\| \cdot \|\mathbf{E}(S_2)\|} \tag{14}$$

where $\mathbf{E}(S)$ represents the BERT embedding of sentence S. Higher scores indicate greater semantic similarity.

Density The density of private information measures how concentrated sensitive personal data (such as names, addresses, etc.) is within a data sample (snippet) using a ratio of recognized private information to the overall length of the snippet where PrivateDataDensity(i) is computed as:

$$\text{Density}(i) = \frac{C_{\text{private}}(i)}{L(i)}$$

- $C_{\text{private}}(i)$ is the count of recognized private information entities (e.g., names, addresses) in snippet i.
- $L(i)$ is the total length of the snippet i (e.g., number of words, characters).

Normalization by the maximum private data density ensures that the density score lies in the range [0, 1]. Higher values indicates that the snippet contains a higher concentration of private information relative to other snippets, while lower values suggest less private data.

Length Length ensures balanced text distribution based on size:

$$\text{Length}(T) = |T| \tag{15}$$

where $|T|$ is the number of words or characters in the text.

Theme Match The similarity is computed using the Kullback-Leibler (KL) divergence based on topic distributions extracted from Latent Dirichlet Allocation (LDA). Specifically, the **ThemeMatch** between snippets i and j is given by:

$$\text{Theme}(i, j) = 1 - \sqrt{\frac{1}{2}\left(D_{\text{KL}}(P_i \| M) + D_{\text{KL}}(P_j \| M)\right)}, \tag{16}$$

where P_i and P_j are the topic distributions of snippets i and j, respectively, and $M = \frac{1}{2}(P_i + P_j)$ is the average topic distribution. The KL divergence D_{KL} measures the divergence between the two topic distributions.

The normalization by max{ThemeMatch} ensures that the theme sensitivity score lies in the range [0, 1], with higher values indicating that snippet i is thematically similar to more snippets in the cluster.

Additional Features We employed additional features to help us select the ones discussed earlier, as not all features consistently yield good results. The effectiveness of a feature heavily depends on the type of text being analyzed and the information we can extract from it. For example, when we tested semantic and textual similarity for biographies, we found that the total unlearning time was higher. This was because biographies are highly unique, making it difficult to identify similarities, such as those based on years or events. However, these similarity measures could be useful in other contexts, such as if a decision was made to remove private data specifically related to "USA citizens," where such similarities would be more effective.

Finally, feature plays a crucial role in refining data distribution strategies. Text and semantic similarity ensure content relevance, density manages cluster balance, length prevents bias, and theme similarity maintains topical consistency. Incorporating these features enhances fairness, efficiency, and reliability in dataset processing and model training.